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Export concentration and diversification impact on economic growth in the developed and developing countries of the world

Abstract. There is much evidence that export diversity has a positive effect on economic growth, but there is some evidence that the concentration of exports may be also related to economic growth. The paper is dedicated to the relationship between export diversification and economic growth in developed and developing countries for 25 years (1996-2020) and 10 years (2011-2020). Correlation between concentration index (Herfindahl-Hirschmann Index), diversification index (modified Finger-Kreinin measure of similarity in trade) and economic growth was studied to confirm or deny the statements concerning positive or negative influence of export diversification and concentration on economic growth.
The results of the study show that concentration index is lower in developed countries and does not varies much over time. While developing countries show higher magnitudes of this index and higher volatility level of this index and its relationship to the world business cycles. Level of export diversification is higher in developed countries, which show more similar export structure to the world pattern comparing with other national economies. Diversification index dynamics in developed and developing countries was getting closer to each other in 2012-2020 and now they are almost the same, though export diversification level of developed countries is a bit closer to the world export pattern comparing to the developing countries index. The findings show that in developing countries export concentration index has sufficient correlation with their GDP growth rates in both periods. That means the more key goods they sell the higher GDP growth

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rate they get. There is no long-time influence on economic development of these states according to this lagged correlation study of 1996-2020 period. Study based on 10-year period show sufficient 1-year lagged correlation rate, but the hypothesis concerning lagged positive impact of export concentration of economic development of these countries during the last decade needs proof. Developed countries have weak or moderate correlation between export concentration and GDP growth. Hence, developing countries mostly concentrate their export on a few primary goods but there is a need of additional research to find out if they use this money to develop other industries. Eventually there are different economies in these groups and some of them have different exports patterns and development models.
Correlation analysis of the relationship between diversification index and GDP growth in 25-year period has shown weak or rather moderate association that does not give us possibility to make an assumption concerning the sufficient impact of export diversification on economic growth. However, in 10-year study we observe high correlation between these data in developing countries, especially in 5-year lag study. If in case of developing countries we observe positive relationship, analysis of developed countries indices shows negative relationship that confirms the statement about role of export diversification in economic growth of developed counties and export concentration - in developing economies. The more their export pattern deviates from the global structure (which is rather diversified) the higher their growth level is. Therefore, their export consists mostly of a few products. Negative relationship between diversification index and GDP growth in developed countries confirms the well-known statement that export diversification should promote economic growth. Lag dynamics in these cases needs additional studies. The list of issues of the following research also includes the search for a model of the nonlinear relationship between export diversification indices and growth rates in economies of different levels of development.
The study has been also focused on the transition economies group, which has shown results more similar to those of developing countries comparing with developed economies. Export concentration index of transition economies is higher than in developing countries and more volatile; export diversification index is much bigger comparing with developing and developed economies. We state that these results reflect existing primary product export strategy of transition economies. The highest concentration and diversification indices are observed in countries that export energy resources.
As a result of the study, the statement that the diversity of exports has a positive effect on economic growth was confirmed. The study has revealed that in developed countries economic growth usually are associated with diversified export basket. The study also showed that over the past 25 years, a developing economy to provide growth did not necessarily need diversified export. Number of countries have shown consistent economic growth driven by export of primary products.
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Вплив концентрації та диверсифікації експорту на економічне зростання

Анотація. Стаття присвячена аналізу взаємозв'язку між диверсифікацією експорту та економічним зростанням у розвинених країнах та країнах, що розвиваються, протягом 25 років (1996-2020) та 10 років (2011-2020). Досліджено взаємозв'язок між індексом концентрації (індекс Герфіндаля-Гіршмана), індексом диверсифікації (модифікований показник подібності торгівлі Фінгера-Крейніна) та економічним зростанням щоб підтвердити або спростувати твердження щодо позитивного чи негативного впливу диверсифікації та концентрації експорту на економічне зростання. У результаті дослідження дістала підтвердження теза щодо того, що різноманітність експорту позитивно впливає на темпи економічного зростання. Дослідження показало, що в розвинених країнах економічне зростання зазвичай пов'язане з різноманітним експортним кошиком. Дослідження також

продемонструвало, що протягом останніх 25 років країнам, щоб забезпечити економічне зростання, не обов'язково було дотримуватися стратегії диверсифікованого експорту. Протягом цього періоду економіка низки країн, що розвиваються та країн з перехідною економікою продемонструвала стійке економічне зростання, зумовлене експортом сировини.
Ключові слова: диверсифікація експорту; концентрація експорту; економічне зростання; кореляція; взаємозв'язок.

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Влияние концентрации и диверсификации экспорта на экономический рост
Аннотация. Статья посвящена анализу взаимосвязи между диверсификацией экспорта и экономическим ростом в развитых и развивающихся странах в течении 25 лет (1996-2020) и 10 лет (2011-2020). Исследована взаимосвязь между индексом концентрации (индекс Герфиндаля-Гиршмана), индексом диверсификации (модифицированный показатель сходства торговли Фингера-Крейнина) и экономическим ростом чтобы подтвердить или опровергнуть утверждение о положительном или отрицательном влиянии диверсификации и концентрации экспорта на экономический рост.
В результате исследования получил подтверждение тезис о том, что разнообразие экспорта положительно влияет на темпы экономического роста. Исследование показало, что в развитых странах экономический рост обычно связан с широкой экспортной корзиной. Исследование также показало, что в течении последних 25 лет странам, чтобы обеспечить экономический рост, не обязательно было придерживаться стратегии диверсифицированного экспорта. В течении этого времени экономика ряда развивающихся стран и стран с переходной экономикой продемонстрировала устойчивый экономический рост, обусловленный экспортом сырья.
Ключевые слова: диверсификация экспорта; концентрация экспорта; экономический рост; корреляция; взаимосвязь.

1. Introduction

The outstanding success of Chinese economy, based on export-focused strategy, demonstrates the role of foreign trade in economic development of a country. Most trade theories state that in the conditions of free trade a country would tend to specialize in certain goods (Bahar, 2016; Sannassee, Seetanah, & Lamport, 2014). Based on Adam Smith's concept of division of labor and specialization for economic growth and development and Heckscher-Ohlin Samuelson's model of international trade, countries should specialize in producing those goods in which they have a comparative advantage. Nonetheless, nowadays it is believed that economic growth and development may be achieved by export diversification (Bahar, 2016; Bida, 2015; Hesse, 2008; Sannassee, Seetanah, & Lamport, 2014). It is argued that countries with commodity concentration are affected by volatility in market prices through changes in foreign exchange revenues. Countries that rely excessively on external demand (as some East Asian countries, before market crash 2007) are more vulnerable to external shocks comparing with those that rely on domestic market (Foxley, 2009), and diversification is the way to decrease negative consequences of dependence from the foreign markets.

There are a lot of evidences that export diversification has a positive influence on the economic growth (Bahar, 2016; Bida, 2015; Hesse, 2008; Hodey, Oduro, & Senadza, 2015; Ivanov, 2018; Sannassee, Seetanah, & Lamport, 2014). There is a number of studies supporting economic development strategy based on export diversification. But there are some evidences that export concentration may be associated with economic growth as well (Basile, Parteka, & Pittiglio, 2015; Alshomaly & Shawaqfeh, 2020). Especially it is relevant for developing countries, characterized by weak domestic demand, therefore exports play sufficient role in growth of their gross domestic product, and in this case export of goods mostly means export of natural raw materials. Considering the end of so-called «primary products business cycle», that provided high level of their GDP growth during the last decades, they need to develop a new export strategy. That's why the issue of export concentration and diversification impact on economic growth is relevant both for these states and the rest of the world.

2. Brief Literature Review

There are numerous studies dedicated to the relationship between foreign trade diversification and economic growth, which state that export diversification contributes to economic growth while export concentration slows down economic growth.

Louis S. Hodey, Abena D. Oduro and Bernardin Senadza (2015) have performed the research on export diversification impact on economic growth using panel data of forty-two Sub-Saharan African countries. Employing the system Generalised Method of Moments estimation technique and three different measures of diversification, they (Hodey, Oduro, & Senadza, 2015) have found that export diversification has a positive and significant effect on economic growth in SSA. The results of the study that has been performed by Caroline Mudenda, Ireen Choga and Cleopas Chigamba (2014) confirm positive impact of export diversification on economic growth of South Africa, that is considered to belong to the group of developing countries. The study (Mudenda, Choga, & Chigamba, 2014) has revealed that export diversification and trade openness are positively related to economic growth in this country. Research on the Chilean experience of the last 30 years performed by S. A. Gutiérrez-de-Piñeres and M. Ferrantino (1997) has revealed that since the mid-1970s, Chilean growth has been accompanied by export diversification, while little diversification had taken place previously. Study performed by Nelson Camanho da Costa Neto and Rafael Romeu (2011) investigates how export industry, destination and product concentration influenced the export performance of Latin American countries during the financial crisis of 2008-2009. The results suggest that product diversification attenuated the trade collapse (Camanho da Costa Neto & Romeu, 2011) contributing to the idea concerning positive impact of export diversification on economy of a country. According to Heiko Hesse's research (2008) (that includes Malaysia, Thailand, Chile and other states), there is a strong evidence that export concentration inhibited economic growth of developing countries in 1961-2000.

As we see number of studies show highly diversified export as a driver of economic growth of a lot of developing countries. But a lot of research results state that high export concentration level has played the key role in economic growth of these countries.

Thus the study of Ibrahim Alshomaly and Walid Shawaqfeh (2020) has found that the exports diversification structures in the group of west Asian Arab countries diverge clearly from the global diversification pattern since the exports of these countries is driven by a high degree of primary exports concentration. Economic growth in the group of countries has been influenced positively by human capital, natural raw materials export growth and the implementation of efficient anti-corruption policies and it has been negatively influenced by trade openness and population growth (Alshomaly & Shawaqfeh, 2020).

Anwesha Aditya and Rajat Acharyya (2011) have performed the study that proves that both statements concerning presence or absence of positive impact of export diversification on economic growth could be correct in certain conditions. They have analyzed data of 65 countries for the period 1965-2005 and taking into consideration other variables like investments, lagged income and infrastructure they have found that both structure and diversification of exports make sufficient impact on economic growth. The authors have found out that specialization and diversification make influence on economic growth depending on export concentration index. They revealed a certain critical level of export concentration above which specialization contributes to higher economic growth, but under that line GDP growth is provided by exports diversification (Aditya & Acharyya, 2011). S. Kadochnikov and A. Fedyunina (2015) state that in general there is the U-shape relationship between foreign trade specialization of a country and its economic development. The results of Heiko Hesse's research (2008) also show that non-linear relationship with developing countries benefiting from diversifying their exports in contrast to the most advanced countries that perform better with export specialization.

Piotr Misztal's (2011) research has found that exports diversification and concentration were among the most important factors that determined the level of gross domestic product per capita in the European Union. The study (Misztal, 2011) has revealed a nonlinear relationship between the degree of exports concentration and gross domestic product per capita in the European Union. The relationship between the degree of exports concentration and GDP per capita took the shape of the letter «W». He calculated the optimum level of export concentration fostering economic growth in EU. Countries in which the degree of exports concentration was higher than optimal level recorded a relatively smaller and positive impact of changes in the exports concentration on GDP per capita (Misztal, 2011).

Non-linear relationship between development level of a state and its export diversification was shown in results of research performed by Deny Bahar (2016). Its data shows that poor countries are highly concentrated, middle income countries are more diversified, and at high levels of development diversification stops and there is a re-specialization pattern. These results are consistent to those published in the paper by Jean Imbs and Romain Wacziarg (2003) who studied the evolution of sectoral concentration in relation to the level of per capita income. They (Imbs & Wacziarg, 2003) show that various measures of sectoral concentration follow a U-shaped pattern across a wide variety of data sources: countries first diversify, in the sense that economic activity is spread more equally across sectors, but there exists, relatively late in the development process, a point at which they start specializing again. Thus, a number of studies show non-linear relationship between export concentration and economic growth.

Number of studies are dedicated to the factors influencing exports concentration and economic growth of different countries. Thus Roberto Basile, Aleksandra Parteka and Rosanna Pittiglio (2015) have performed research of 114 countries' (1992-2012) exports data. The estimation results corresponds with theoretic basis of endogenous growth models: developed states export more goods because of higher competitiveness of their enterprises due to mainly technological advantages in the world markets, while large countries (e.g. China) use advantages of economies of scale and use lower factor costs to compensate lower fundamental productivity. On the contrary, technology transfer contribute to higher productivity in these countries and allow them to sell abroad a wide basket of goods. Moreover, good geographic position next to developed countries helps the diversification process, because in past decades developed countries tended to transfer their industrial facilities abroad to enjoy lower factor costs of their neighbours (Basile, Parteka, & Pittiglio 2015). However, with the collapse of the mortgage market in 2007, the period of economic growth based on cheap exports came to an end for many countries and they faced the challenges of a global recession. Therefore, today one can observe the following trend: multinational companies are moving their industrial facilities back to the territory of their home countries. This is happening because of development of automation and robotization in manufacturing, increasing the share of high-tech products in the consumer basket, and, consequently, reducing the share of production that requires a lot of cheap labor or large amounts of raw materials.

Marlo Murphy-Braynen (2019) has performed the research concerning factors that make impact on export diversification and economic growth. According to the study (Murphy-Braynen, 2019) the key factors that support export diversification are domestic investment, quality of institutions, population, human capital, high level of education, quality of infrastructure and market access. These conditions are typical for developed countries, which have a higher level of export diversification compared to developing countries. Among the factors that increase the concentration of exports, Murphy-Brainan (2019) highlights: large economic distance from big markets, trade liberalization, declining terms of trade, high volatility of exchange rate and its overvaluation etc. (Murphy-Braynen, 2019). The combination of raw commodity specialization with periodic devaluations of the national currency is typical of a number of post-socialist economies.

Muhammad Tariq Majeed and Eatzaz Ahmad (2006) also have found a significant role of exchange rate policy for both the structure of exports and for the economic development of the country in general. Based on data on 75 developing countries over the period 1970-2004 they have performed the study to summarize the main factors that promote exports in these countries. They revealed that development of the net of communication facilities is crucial not only in promoting economic growth, as is well known, it is also important for sustained exports performance. The researchers state that a stable exchange rate policy is determinative for avoiding risks associated with the assets, import prices and profit considerations of direct investor in developing countries (Majeed & Ahmad, 2006). After all, the undervalued national currency means excess profits of exporters of raw materials, and, accordingly, the lack of motivation to produce / export high-tech products. At the same time, the import of equipment, which can be used to produce high-tech goods, is becoming more expensive.

Whilst numerous studies are dedicated to impact of exports structure on economic growth of different countries final verdict on the relationship between export concentration and diversification and economic growth in developed and developing countries has remained inconclusive and the role of export diversification in their economic development still needs additional researches. There are three sets of studies: one of them proves that export concentration slows down economic growth, the second one analyses the cases when economic growth was driven by highly concentrated export and shows that export diversification is not high export diversification level is not mandatory for high economic growth and the third one states that influence of export concentration or diversification level on economic growth, it is not determinant and it depends on certain circumstances. The aim of this study is to contribute to one of them.

3. Purpose

The purpose of the research is to study the relationship between export diversification and economic growth in developed and developing countries for 25 years (1996-2020) and 10 years (2011-2020).

The tasks of the study are the following:

- to analyze the correlation between GDP per capita growth dynamics (annual %) and Herfindahl-Hirschmann Index for 0-5 year lags;
- to analyze the correlation between GDP per capita growth dynamics (annual %) and diversification index (modified Finger-Kreinin measure of similarity in trade) for 0-5 year lags;
- to find out if export concentration and diversification makes positive or negative impact on economic growth in developed and developing countries.

4. Methodology and Data

The following indices, computed by United Nations Conference on Trade and Development (2021), have been used to evaluate the level of foreign trade diversification of a country and its similarity to the world trade structure:

- ***concentration index***, also named Herfindahl-Hirschmann Index (HHI), is a measure of the degree of product concentration. Normalized HHI obtains values between 0 and 1 (United Nations Conference on Trade and Development, 2021):

$$H_j = \frac{\sqrt{\sum_{i=1}^{n}\left(\frac{x_{ij}}{X_j}\right)^2} - \sqrt{1/n}}{1-\sqrt{1/n}}, \tag{1}$$

where:
H_j - country or country group index;
x_{ij} - value of export for country j and product i;
n - number of products;
$X_j = \sum_{i=1}^{n} x_{ij}$.

An index value closer to 1 indicates that a country's exports or imports are highly concentrated on a few products. On the contrary, values closer to 0 reflect exports or imports are more homogeneously distributed among a series of products (United Nations Conference on Trade and Development, 2021).

- ***diversification index*** (modified Finger-Kreinin measure of similarity in trade) is computed by measuring the absolute deviation of the trade structure of a country from the world structure (United Nations Conference on Trade and Development, 2021):

$$S_j = \frac{\sum_i |h_{ij} - h_i|}{2}, \tag{2}$$

where:
h_{ij} - share of product i in total exports or imports of country or country group j;
h_i - share of product i in total world exports or imports.

The diversification index takes values between 0 and 1. A value closer to 1 indicates greater divergence from the world pattern (United Nations Conference on Trade and Development, 2021). That's why in this study high rate of diversification index means a narrow export basket, that is a deep contrast to the world export structure, that is rather wide. Nevertheless, in literature above highly diversified export structure means selling abroad a long list of goods.

In the study the Pearson's correlation test was used to analyze the relationship between GDP growth and selected indices. Calculations were performed by means of Microsoft Excel. Data was collected from United Nations Conference on Trade and Development statistic data for the period 1996-2020 (United Nations Conference on Trade and Development, 2021).

5. Results

The research results show that export diversification in developed and developing economies differ from each other. The results of the study show that concentration index is lower in developed countries and does not varies much over time (Figure 1). Export concentration is higher in developing states. Another worth mentioning fact is that in developing countries one can observe higher volatility level of this index and its relationship to the world business cycles. There are obvious periods of concentration growth during the world economy booms and sharp declines of this index after stock crashes bringing the post Dot-com bubble crisis (2001) and Subprime mortgage crisis in US (2007). Such dynamics of the concentration index confirms the above-mentioned thesis on the diversity of the export basket as one of the elements of economic stability of the state.

The results of the study of diversification index show that its dynamics in developed and developing countries is getting more similar especially since 2012 (Figure 1). Nowadays export diversification level of developed countries is a bit closer to the world export pattern comparing to the developing countries index.

The results of the study of the correlation between the indices of concentration and diversification of exports and the dynamics of GDP are shown in Table 1. Analysis of data from the period 1996-2020 show that there is rather high and positive correlation between concentration index and GDP per capita of developing countries with 0-1 year time lag, but the bigger time lag the less correlation magnitude we observe (Table 1 and Figure 2). Correlation analysis for developing countries based on data from period 2011-2020 show almost the same correlation level with 0 lag, we observe sufficient growth of the correlation level 1-year lags, declining for 2 and 3-lag period and sharp fall reaching negative magnitudes for 4 and 5-lag period. The matter of fact this positive relationship means the more key goods they export the higher GDP growth rate they have. These results show that there is the lagged impact on economic development made by that export like other industries or infrastructure. The strongest lagged influence is observed for 1-lag period and

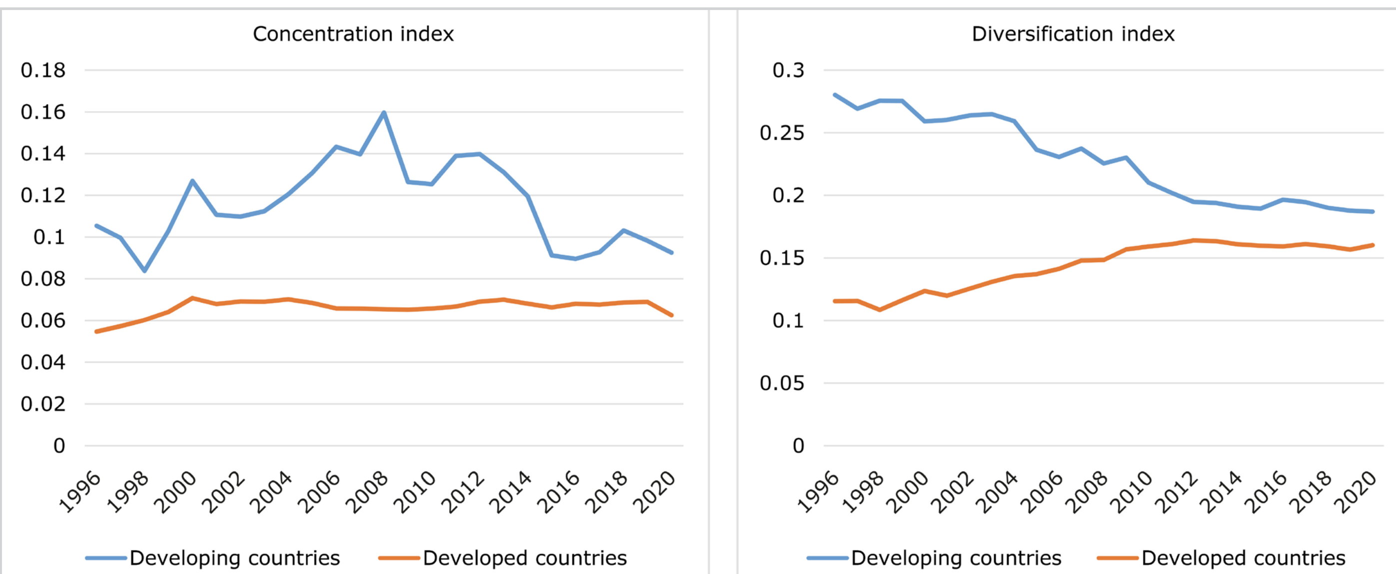

Figure 1:
Export concentration and diversification indices of developed and developing countries in 1996-2020
Source: Developed by the authors, 2021

Table 1:
Correlation between concentration and diversification indices on one side and GDP per capita growth rates on the other in developed and developing countries in 1996-2020 and in 2011-2020 (lag equals 0-5 years)

Indices	Lag, years	Number of observations, years	Developing countries	Developed countries
1996-2020				
concentration index	0	25	0.597434	0.056576
	1	24	0.634009	-0.04177
	2	23	0.432702	-0.04154
	3	22	0.350622	-0.12636
	4	21	0.29851	-0.14039
	5	20	0.055123	-0.06842
diversification index	0	25	0.143594	-0.4309
	1	24	-0.07725	-0.47324
	2	23	-0.11243	-0.46277
	3	22	-0.16111	-0.47654
	4	21	-0.25244	-0.49323
	5	20	-0.35007	-0.42097
2011-2020				
concentration index	0	10	0.515687	0.773312
	1	9	0.883123	-0.04517
	2	8	0.810128	-0.25483
	3	7	0.492763	-0.40019
	4	6	-0.49589	0.192712
	5	5	-0.4025	-0.65374
diversification index	0	10	0.639779	-0.04439
	1	9	0.449199	-0.77274
	2	8	0.222443	-0.40927
	3	7	0.180271	0.176823
	4	6	0.437154	-0.72044
	5	5	0.959863	-0.35419

Source: Calculated by the authors, 2021

declines for longer lags. This fact let us assume that these countries use export of primary products for further economic development.

In the second half of 90s and up to 2008 developing countries gained benefits from so-called «primary products business cycle», selling abroad mostly natural raw materials and demonstrating high GDP growth rates. After mortgage market crash in 2007 their growth rates decreased as their export concentration, which once again proves the thesis of reducing the efficiency of the raw material model of growth in modern conditions. But it should be mentioned that since 2007 GDP growth rates of developed countries have fallen too, but their export concentration hasn't changed, that may be explained by their non-primary products type of economy.

The research of 25-year data show that in developed countries there is almost no correlation between concentration index and GDP per capita - correlation level in most cases is less than 0.15. But 10-year study (2011-2020) brings another results. 0-lag correlation between their export concentration and GDP growth equals 0.773312, but sufficiently declines for 1-5 lag periods reaching negative magnitudes. This strong 0-lag relationship for developed countries may be explained by the fact that since 2020 UNCTAD has included some post-socialist countries (like Ukraine and Russian Federation which are strongly oriented on export of raw materials) to the group of developed countries, that could influence the result of the whole group of developed countries. The previous research based on 2000-2019 data (where post-socialist countries have been included to the separate transition group and therefore have made no impact on developed countries aggregate data) has shown no substantial relationship between concentration and GDP growth in developed countries.

Analysis of 25-year data shows that correlation between diversification level and GDP growth in developing countries is mostly negative and it grows as time lad increases, but the relationship is too weak to jump into conclusions. 10-year data study brings other results. In this case correlation is positive, it's middle in case of 0 lag, its magnitude decreases as the lag grows up to 3-year lag period and after that sufficiently grows in case of using 4 and 5-year lags reaching in the last case 0.959863. That means developing countries get benefits from their export structure deviation from the global pattern and we've got another confirmation of great dependence of GDP growth in developing countries from exports of primary products. High concentration and diversification indices are observed in countries that export energy resources like Azerbaijan and Turkmenistan.

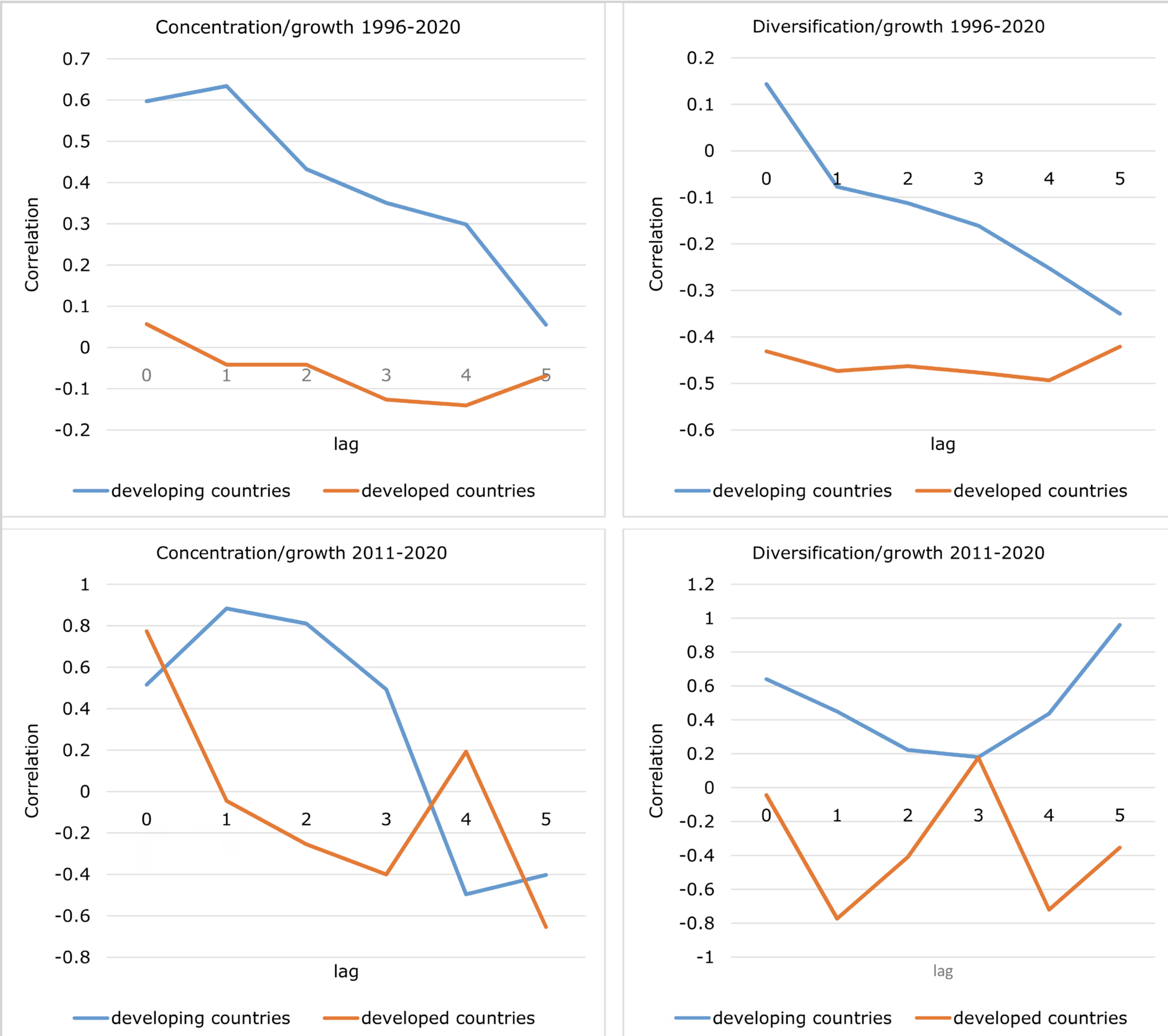

Figure 2:
Correlation between concentration and diversification indices on one side and GDP per capita growth rates on the other in developed and developing countries in 1996-2020 and in 2011-2020 (lag equals 0-5 years)
Source: Developed by the authors, 2021

In case of developed countries correlation between diversification level and economic growth is negative. In case of 25-year study the relationship is indirect and weak that doesn't give us an opportunity to jump into conclusions. But in 10-year data research relationship between Diversification index dynamics and GDP growth rates is indirect and reaches up to -0.77274 in case of 1-year lag correlation, downsizes in 2-year and 3-year lag cases and grows in 4-year lag study reaching -0.77274. Basically negative correlation in this case means the more export structure of developed countries is similar to the world pattern the higher their GDP growth is. 1-year and 4-year lagged impact in this case needs additional researches.

As Ukraine has recently been included into the group of so-called transitional economics, we have decided focus this research on this group as well. Our study has been based on 1995-2019 data because since 2020 UNCTAD has not calculated aggregate indices for so-called transition countries. In general for this group we observed results that are more similar to those of developing countries comparing with results of developed countries. Export concentration index of transition economies is higher than in developing countries and more volatile; export diversification index is much bigger comparing with developing and developed economies. We state that these results reflect existing primary product export strategy of transition economies. Unfortunately, post-socialist countries still focus on the export of mainly raw materials and low-tech goods, which is accompanied by all the negative consequences of such policies - significant dependence on global business cycles, world commodity markets conditions, undervalued national currency etc. The highest concentration and diversification indices are observed in countries that export energy resources like Azerbaijan and Turkmenistan. Oil and gas exporting countries show the highest levels of GDP growth in the group of transitions states.

Results of the study contribute to the general idea about positive influence of export diversification on economic growth as the study shows that developed countries have low export concentration index and the group of developing countries vice versa. Thus the research results contribute to the study performed by Bahar (2016), Bida (2015), Hesse (2008), Hodey, Oduro, and Senadza (2015), Ivanov (2018), Sannassee, Seetanah and Lamport et al. (2014). Our findings reveal that in the last two and a half decades (especially in 1995-2007) in developing countries economic growth was caused mostly by primary products exports. This research has also confirmed the statement expressed by Ibrahim Alshomaly and Walid Shawaqfeh (2020) concerning West Asian Arab states that happens to be oil exporters. Also our findings correspond with H. Hesse (2008), who believes that while developing countries benefit from export diversification, developed countries achieve economic growth through export specialization.

6. Conclusions and Perspectives for Further Discussion

Nowadays developed states use export diversification benefits more widely comparing with developing economies that often depend on a narrow export basket. There are still a lot of countries that rely on traditional export structure including primary products, that being combined with constant devaluation of their national currencies, leads to enrichment of a certain business groups and sealing obsolete economy structure of these states.

The results of the study show that concentration index is lower in developed countries and does not varies much over time. While developing countries show higher magnitudes of this index and higher volatility level of this index and its relationship to the world business cycles. Level of export diversification is higher in developed countries, which show more similar export structure to the world pattern comparing with other national economies. Diversification index dynamics in developed and developing countries was getting closer to each other in 2012-2020 and now they are almost the same, though export diversification level of developed countries is a bit closer to the world export pattern comparing to the developing countries index.

The correlation analysis has been performed using the data from two periods: 1996-2020 and 2011-2020. The second period has been studied separately to eliminate the influence of the so-called «primary products business cycle» and to analyze export concentration and diversification when the global economic turbulence caused by market crash in 2007 has declined.

The findings show that in developing countries export concentration index has sufficient correlation with their GDP growth rates in both periods. That means the more key goods they sell the higher GDP growth rate they get. There is no long-time influence on economic development of these states according to this lagged correlation study of 1996-2020 period. Study based on 10-year period show sufficient 1-year lagged correlation rate, but the hypothesis concerning lagged positive impact of export concentration of economic development of these countries during the last decade needs proof. Developed countries have weak or moderate correlation between export concentration and GDP growth. So developing countries mostly concentrate their export on a few primary goods but there is a need of additional research to find out if they use this money to develop other industries. Eventually there are different economies in these groups and some of them have different exports patterns and development models.

Correlation analysis of the relationship between diversification index and GDP growth in 25-year period has shown weak or rather moderate association that does not give us possibility to make an assumption concerning the sufficient impact of export diversification on economic growth. But in 10-year study we observe high correlation between these data in developing countries, especially in 5-year lag study. If in case of developing countries we observe positive relationship, analysis of developed countries indices show negative relationship that confirms the statement about role of export diversification in economic growth of developed countries and export concentration - in developing economies. The more export pattern of developing countries deviate from the global structure (which is rather diversified) the higher their growth level is. Therefore their export consists mostly of a few products. Negative relationship between diversification index and GDP growth in developed countries confirms the well-known statement that export diversification should promote economic growth. Lag dynamics in these cases needs additional studies. The list of issues of the following research also includes the search for a model of the nonlinear relationship between export diversification indices and growth rates in economies of different levels of development.

The study has been also focused on the transition economies group, which has shown results more similar to those of developing countries comparing with developed economies. Export concentration index of transition economies is higher than in developing countries and more volatile; export diversification index is much bigger comparing with developing and developed economies. We state that these results reflect existing primary product export strategy of transition economies. Unfortunately, post-socialist countries still focus on the export of mainly raw materials and low-tech goods, which is accompanied by all the negative consequences of such policies - significant dependence on global business cycles, world commodity markets conditions, undervalued national currency etc. The highest concentration and diversification indices are observed in countries that export energy resources like Azerbaijan and Turkmenistan. Oil and gas exporting countries show the highest levels of GDP growth in the group of transitions states.

The novelty of the paper lies in the following. The study revealed that during the last 25 years a country to provide economic growth did not necessarily need diversified export. But the long-run effect provided by primary products export-led growth remains questionable. The more important thing that the decade of primary-led export growth has come to an end that raises the question concerning the effect of this growth model in the future. But export diversification itself may not be an effective growth model for developing countries, because companies from developed countries do not transfer their production facilities overseas en masse as we observed during last decades. Today, the opposite trend is observed - the transfer of production capacity back to the home countries, which began as a result of the global recession and became possible due to the increasing degree of automation of production processes. To ensure high or at least moderate economic growth, developing countries need to develop a new modern growth model that could be a topic for further research.

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